Savitribai Phule Pune University (SPPU) | BE Artificial Intelligence & Data Science

MACHINE LEARNING – UNIT-WISE COMPREHENSIVE VIVA GUIDE

Course Code: 414401 | Academic Year 2020 Pattern Syllabus & Laboratory-I Mapping

UNIT 1 – Introduction to Machine Learning

Q1. What is Machine Learning?

Answer: Machine Learning (ML) is a branch of Artificial Intelligence (AI) and computer science that focuses on building systems that learn from data, identify patterns, and make decisions or predictions with minimal human intervention. As defined by Tom Mitchell: *"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."*

Q2. What is Artificial Intelligence?

Answer: Artificial Intelligence (AI) is a broad multidisciplinary field of computer science aimed at creating smart machines capable of performing tasks that typically require human intelligence. These tasks include logical reasoning, problem-solving, natural language understanding, visual perception, decision-making, and autonomous navigation.

Q3. What is the difference between AI, ML, and Deep Learning?

Answer:

- **Artificial Intelligence (AI):** The overarching superset aimed at creating intelligent systems that simulate human intelligence.
- **Machine Learning (ML):** A subset of AI where statistical algorithms learn patterns directly from structured or unstructured data without being explicitly programmed with hardcoded rules.
- **Deep Learning (DL):** A specialized subset of ML based on multi-layered Artificial Neural Networks (ANNs) designed to extract high-level representations from massive, complex datasets.

Q4. What is Data Science?

Answer: Data Science is an interdisciplinary field that combines statistics, data analysis, domain expertise, computer engineering, and machine learning to extract actionable knowledge and meaningful insights from large-scale structured, semi-structured, and unstructured datasets.

Q5. What are the main types of Machine Learning?

Answer:

1. **Supervised Learning:** Models are trained using historical datasets containing input features paired with ground-truth target labels.
2. **Unsupervised Learning:** Algorithms explore unlabeled data to discover hidden patterns, underlying structures, or natural groupings.
3. **Semi-Supervised Learning:** Combines a small amount of labeled data with a large amount of unlabeled data during training to improve accuracy while reducing labeling costs.
4. **Reinforcement Learning:** An autonomous agent learns optimal decision policies through iterative trial-and-error interactions with an environment, guided by scalar rewards and penalties.

Q6. What is Supervised Learning? Give examples.

Answer: Supervised learning is an ML paradigm where the model learns a mapping function $f(X) = Y$ between input features X and ground-truth output labels Y. The model minimizes prediction errors by comparing its estimates with the actual target labels.

Examples: Spam email detection (Classification), Real-estate house price prediction (Regression), Medical diagnostic classification.

Q7. What is Unsupervised Learning? Give examples.

Answer: Unsupervised learning processes unlabeled input datasets (X) without corresponding target answers (Y). The goal is to uncover hidden structure, probability density, latent representations, or intrinsic data groupings.

Examples: E-commerce customer segmentation (K-Means), Anomaly detection in financial fraud, High-dimensional visualization using PCA.

Q8. What is Semi-Supervised Learning and where is it used?

Answer: Semi-Supervised Learning leverages a tiny fraction of labeled data alongside a substantial volume of unlabeled data. It is widely used in real-world scenarios where manual data labeling is prohibitively expensive, time-consuming, or requires domain experts.

Applications: Medical image analysis (e.g., MRI scans where annotation by radiologists is scarce), Speech recognition, Web content classification.

Q9. What is Reinforcement Learning?

Answer: Reinforcement Learning (RL) is a paradigm where an autonomous agent operates in a dynamic environment, observing states, executing actions, and receiving numerical feedback (rewards or penalties). The agent's goal is to maximize cumulative long-term expected reward by mastering an optimal behavioral policy.

Q10. Compare Classification vs. Regression.

Feature	Classification	Regression
Target Variable	Discrete categorical labels (e.g., Spam/Ham, Class A/B/C)	Continuous numerical values (e.g., Price, Salary, Temperature)
Output Goal	Find decision boundary separating classes	Fit a best-fit curve/line estimating relationship
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC	Mean Absolute Error (MAE), MSE, RMSE, R^2 Score

Q11. Define Dataset, Feature, and Label with an example.

Answer:

- **Dataset:** A structured collection of data points (rows/observations) and dimensions (columns/attributes).
- **Feature (X):** An individual measurable property or independent input variable used by the model for learning (e.g., Square Footage, Number of Bedrooms, Location score).
- **Label (Y):** The target dependent variable or ground-truth output that the model aims to predict (e.g., House Price = $350,000).

Q12. What is Principal Component Analysis (PCA)?

Answer: PCA is an unsupervised, linear dimensionality reduction technique that transforms high-dimensional features into a lower-dimensional subspace of orthogonal, uncorrelated features called Principal Components (PCs). It minimizes information loss by maximizing variance along orthogonal eigenvectors calculated from the data covariance matrix.

Q13. What is Linear Discriminant Analysis (LDA)?

Answer: LDA is a supervised dimensionality reduction and classification technique. Unlike PCA (which maximizes total variance), LDA maximizes class separability by finding a linear combination of features that maximizes the between-class variance relative to the within-class variance.

Q14. Explain Overfitting and Underfitting with remedies.

Answer:

- **Overfitting (High Variance):** Occurs when a model learns the noise and exact details of the training data rather than underlying patterns, performing exceptionally on training data but failing on unseen test data. *Remedies:* Add L1/L2 regularization, reduce model complexity, perform cross-validation, apply dropout, or increase training data.
- **Underfitting (High Bias):** Occurs when a model is overly simplistic to capture underlying trends in data, performing poorly on both training and test sets. *Remedies:* Increase model capacity/complexity, add engineered polynomial features, reduce regularization strength.

UNIT 2 – Regression Analysis

Q1. What is Linear Regression and its underlying assumptions?

Answer: Linear Regression is a supervised learning algorithm that models the linear relationship between a dependent scalar variable *y* and one or more explanatory independent variables *x*. Key assumptions include:

1. Linearity between inputs and target.
2. Homoscedasticity (constant error variance across predictors).
3. Multicollinearity absence (independent features are not highly correlated).
4. Independence and normality of residual error terms.

Q2. State the equations for Simple and Multiple Linear Regression.

Answer:

> $\text{Simple Linear Regression: } y = \beta_0 + \beta_1 x + \varepsilon$
> $\text{Multiple Linear Regression: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$

Where β_0 is the y-intercept, $\beta_1 \ldots \beta_p$ are feature slope coefficients, and ε represents the random residual error term.

Q3. What is the difference between Regression and Correlation?

Answer: Correlation quantifies the direction and strength of a linear association between two random variables (bounded between -1 and +1), treating both symmetrically without assuming causation. Regression establishes a functional mathematical mapping to predict a dependent response variable based on independent input predictors.

Q4. Differentiate between Ridge (L2) and Lasso (L1) Regularization.

Property	Ridge Regression (L2)	Lasso Regression (L1)
Penalty Term	Penalty = $\lambda \sum (\beta_j)^2$ (Squared Magnitude)	Penalty = $\lambda \sum \lvert\beta_j\rvert$ (Absolute Value)
Feature Selection	Shrinks coefficients near zero, retains all features.	Drives uninformative coefficients strictly to zero (Sparse Models).
Best Used For	Handling severe multicollinearity among features.	Automatic feature selection in high-dimensional datasets.

Q5. What is ElasticNet Regression?

Answer: ElasticNet is a regularized linear regression model that linearly combines both L1 (Lasso) and L2 (Ridge) penalty terms in its cost function:

```
Cost = MSE + λ₁ ∑ |βⱼ| + λ₂ ∑ (βⱼ)²
```

It overcomes Lasso's limitation of selecting at most *n* features when *p* > *n* and performs well when dealing with multiple correlated feature groups.

Q6. What is Polynomial Regression?

Answer: Polynomial Regression models non-linear relationships by raising independent feature variables to specified polynomial degrees (e.g., $y = \beta_0 + \beta_1 x + \beta_2 x^2 + ... + \beta_n x^n$). Although it fits non-linear curves, it is still categorized as linear regression because the regression parameters (β) remain linear.

Q7. Explain the Bias-Variance Tradeoff mathematically and conceptually.

Answer: Total Expected Generalization Error is decomposed into three components:

```
Total Error = (Bias)² + Variance + Irreducible Error (σ²)
```

- **Bias:** Error introduced by approximating a complex real-world problem with an oversimplified model (causes underfitting).
- **Variance:** Error stemming from sensitivity to small fluctuations/noise in the training dataset (causes overfitting).

The goal is to find optimal hyperparameter settings that balance low bias with low variance.

Q8. Detail the key evaluation metrics for Regression models.

- **MAE (Mean Absolute Error):** Average of absolute prediction differences: `MAE = (1/n) ∑ |yᵢ - ŷᵢ|`. Robust against extreme outliers.
- **MSE (Mean Squared Error):** Average of squared errors: `MSE = (1/n) ∑ (yᵢ - ŷᵢ)²`. Heavily penalizes large outlier errors.
- **RMSE (Root Mean Squared Error):** Square root of MSE: `RMSE = √(MSE)`. Expressed in original target units.
- **R² Score (Coefficient of Determination):** Represents the proportion of target variance explained by inputs relative to a baseline mean model: `R² = 1 - (SS_res / SS_tot)`.
- **Adjusted R²:** Adjusts R² to account for the number of predictors (*p*) added to the model: `Adj R² = 1 - [(1 - R²)(n - 1) / (n - p - 1)]`. Prevents artificial R² inflation.

UNIT 3 – Classification Techniques

Q1. Explain Logistic Regression. Why is it used for classification?

Answer: Despite its name, Logistic Regression is a linear classification algorithm used for predicting categorical probabilities. It applies the non-linear Sigmoid (logistic) function to transform a linear log-odds combination of features into a probability value bounded strictly between 0 and 1:

```
σ(z) = 1 / (1 + e⁻ᶻ), where z = wᵀx + b
```

If σ(z) ≥ 0.5, the instance is classified into the positive binary class.

Q2. How does K-Nearest Neighbors (KNN) classify a data point? What is the impact of K?

Answer: KNN is a non-parametric, instance-based "lazy learner." It stores all historical training instances without building an explicit internal model. To classify a query point, it calculates distance metrics (e.g., Euclidean or Manhattan) to find the *K* nearest training samples and assigns the class label via majority voting.

- Small K values (e.g., K=1) lead to high variance, noise sensitivity, and overfitting.
- Large K values smooth out boundaries, leading to high bias and underfitting.

Q3. Explain Support Vector Machines (SVM) along with Hyperplane, Margin, and Support Vectors.

Answer: SVM finds an optimal linear decision boundary (Hyperplane) in feature space that maximizes the geometric distance (Margin) between support vectors belonging to opposite classes.

- **Hyperplane:** A decision boundary defined as $w^T x + b = 0$ that separates distinct classes.
- **Support Vectors:** Critical data points positioned nearest to the decision hyperplane that define its location and orientation.
- **Margin:** The distance between the separating hyperplane and the closest support vector (equal to $2 / ||w||$). SVM maximizes this margin.

Q4. What is the Kernel Trick in SVM? List popular kernels.

Answer: The Kernel Trick enables SVM to operate in high-dimensional feature spaces without explicitly calculating expensive coordinate transformations. A Kernel function computes inner dot products of data points directly in a higher-dimensional implicit feature space where non-linearly separable data becomes linearly separable.

Popular Kernels: Linear, Polynomial, Radial Basis Function (RBF / Gaussian), Sigmoid.

Q5. Explain the Confusion Matrix and derive Precision, Recall, Accuracy, and F1-Score.

Answer: A Confusion Matrix evaluates binary/multiclass classification performance:

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN) - Type II Error
Actual Negative	False Positive (FP) - Type I Error	True Negative (TN)

```
Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP) [Focuses on minimizing False Positives]
Recall (Sensitivity) = TP / (TP + FN) [Focuses on minimizing False Negatives]
F1-Score = 2 × (Precision × Recall) / (Precision + Recall) [Harmonic Mean]
```

Q6. Compare One-vs-Rest (OvR) and One-vs-One (OvO) strategies for Multiclass Classification.

- **One-vs-Rest (OvR):** Fits N individual binary classifiers for an N-class problem. Each model trains on one class vs. all remaining combined classes. Fast and scalable.
- **One-vs-One (OvO):** Fits $N(N - 1)/2$ binary models evaluating every distinct pairwise combination of classes. Computationally expensive for large class counts, but works well with algorithms that do not scale well to large datasets (e.g., SVMs).

UNIT 4 – Unsupervised Learning & Clustering

Q1. Explain the K-Means Clustering algorithm step-by-step.

1. **Initialization:** Randomly choose K initial cluster centroids in feature space.
2. **Assignment Step:** Compute Euclidean distances and assign each data point to its nearest centroid.
3. **Update Step:** Recalculate each centroid as the mean coordinate vector of all points assigned to that cluster.
4. **Convergence:** Repeat Assignment and Update steps until centroid coordinates stabilize or maximum iterations are reached.

Q2. How do you select the optimal K using Elbow Method and Silhouette Analysis?

- **Elbow Method:** Plots Within-Cluster Sum of Squares (WCSS / Inertia) against a range of *K* values. The "elbow point" on the curve indicates the optimal tradeoff between variance explained and cluster count.
- **Silhouette Score:** Quantifies how well-separated clusters are. Measures cohesion *a(i)* (mean intra-cluster distance) and separation *b(i)* (mean nearest-cluster distance):

  ```
  s(i) = [b(i) - a(i)] / max(a(i), b(i))
  ```

  Values range from -1 (poor clustering) to +1 (dense, well-separated clusters).

Q3. Compare Agglomerative and Divisive Hierarchical Clustering.

- **Agglomerative (Bottom-Up):** Begins with each individual data point in its own distinct cluster and recursively merges nearest pairs based on linkage criteria (Single, Complete, Average, Ward) until a single cluster remains. Visualized via a Dendrogram.
- **Divisive (Top-Down):** Begins with all data instances inside a single unified cluster and recursively splits heterogeneous clusters until each data point forms an isolated cluster.

Q4. How does DBSCAN work? Define Core, Border, and Noise points.

Answer: Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clusters points based on local point density rather than distance from centroids, allowing it to discover arbitrarily shaped clusters and isolate noise outliers. Key hyperparameters: *Eps (ε)* (neighborhood radius) and *MinPts* (minimum density threshold).

- **Core Point:** Contains at least *MinPts* within its *Eps*-neighborhood.
- **Border Point:** Contains fewer than *MinPts* within *Eps*, but falls within the *Eps*-neighborhood of a Core Point.
- **Noise Point:** Any data point that is neither a Core Point nor a Border Point.

Q5. Differentiate between K-Means, K-Medoids, and K-Prototype.

- **K-Means:** Uses mean vectors as centroids; sensitive to extreme noise and outliers.
- **K-Medoids (PAM):** Uses actual observed data points as cluster representatives; robust against noise and outliers.
- **K-Prototype:** Integrates K-Means (for numerical features) and K-Modes (for categorical features) to handle mixed-attribute datasets.

UNIT 5 – Ensemble Learning Methods

Q1. What is Ensemble Learning? What are its primary advantages?

Answer: Ensemble Learning combines predictions from multiple base models (weak learners) to construct a stronger predictive system. It improves overall generalization performance, reduces prediction variance, and minimizes model bias compared to individual standalone models.

Q2. Explain Bagging (Bootstrap Aggregating) and Random Forest.

Answer:

- **Bagging:** Generates multiple random training subsets using bootstrap sampling (sampling with replacement). Fits independent base estimators in parallel and aggregates predictions via majority voting (classification) or averaging (regression).
- **Random Forest:** An extension of Bagging applied to Decision Trees. Introduces feature randomness by evaluating only a random subset of features at each node split, decorrelating individual decision trees and reducing overall variance.

Q3. Explain Boosting and contrast AdaBoost, Gradient Boosting, and XGBoost.

Answer: Boosting trains base estimators sequentially, where each subsequent model focuses on correcting the errors made by preceding models.

- **AdaBoost:** Increases weight multipliers for misclassified instances after each round, forcing subsequent weak decision stumps to focus on difficult cases.
- **Gradient Boosting (GBM):** Iteratively fits new base learners to the negative gradient (pseudo-residuals) of a loss function using gradient descent.
- **XGBoost:** An optimized implementation of Gradient Boosting featuring built-in L1/L2 regularization, tree pruning, parallelized split detection, and native handling of missing values.

Q4. Compare Stacking, Blending, and Voting ensemble techniques.

- **Voting:** Combines predictions from multiple distinct base estimators using simple majority rules (Hard Voting) or weighted average probabilities (Soft Voting).
- **Stacking:** Trains multiple base estimators (Level 0) and passes their prediction outputs as feature inputs to a high-level Meta-Model (Level 1) trained using cross-validation.
- **Blending:** Similar to Stacking, but derives meta-model training features from a fixed holdout validation dataset rather than out-of-fold cross-validation.

UNIT 6 – Reinforcement Learning

Q1. Define key RL components: Agent, Environment, State, Action, Reward, Policy, Value Function.

- **Agent:** The autonomous decision-maker that interacts with the environment.
- **Environment:** The external system or world observed and acted upon by the agent.
- **State (s):** Complete representation of the environment at time step t.
- **Action (a):** Possible decisions or moves executed by the agent.
- **Reward (r):** Scalar feedback signal evaluated by the environment assessing action quality.
- **Policy (π):** The agent's strategy mapping states to actions: $a = \pi(s)$.
- **Value Function V(s):** Expected cumulative discounted reward starting from state s.

Q2. What is Markov Decision Process (MDP)? Explain the Markov Property.

Answer: MDP provides a mathematical framework for modeling decision-making under uncertainty, defined by the tuple (S, A, P, R, γ). It relies on the **Markov Property**: *"The future state depends only upon the current state and action, independent of the past historical trajectory."*

$$P(S_{t+1} \mid S_t, A_t, S_{t-1}, A_{t-1}, \ldots) = P(S_{t+1} \mid S_t, A_t)$$

Q3. State and explain the Bellman Expectation Equation.

Answer: The Bellman Equation decomposes value function computations recursively into the immediate reward plus the discounted value of the subsequent successor state:

$$V(s) = E\left[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s \right]$$

Where $\gamma \in [0, 1)$ represents the discount factor prioritizing immediate vs. long-term future rewards.

Q4. Explain Q-Learning and state the Q-Value update equation.

Answer: Q-Learning is a model-free, off-policy Temporal Difference (TD) reinforcement learning algorithm designed to learn the optimal action-value function $Q^*(s, a)$ without prior knowledge of environment transition dynamics. The Q-table update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Where α is the learning rate, r is immediate reward, γ is discount factor, and $\max_{a'} Q(s', a')$ estimates maximum future expected reward from next state s'.

Q5. Explain the Exploration vs. Exploitation Tradeoff and ε-Greedy strategy.

Answer:

- **Exploration:** Trying random or unfamiliar actions to gather information about the environment.
- **Exploitation:** Selecting best-known actions with highest estimated rewards to maximize immediate payoff.
- **ε-Greedy Strategy:** Selects a random exploratory action with probability ε, and selects the greedy optimal action with probability $(1 - \varepsilon)$. ε decays over time as the agent learns.